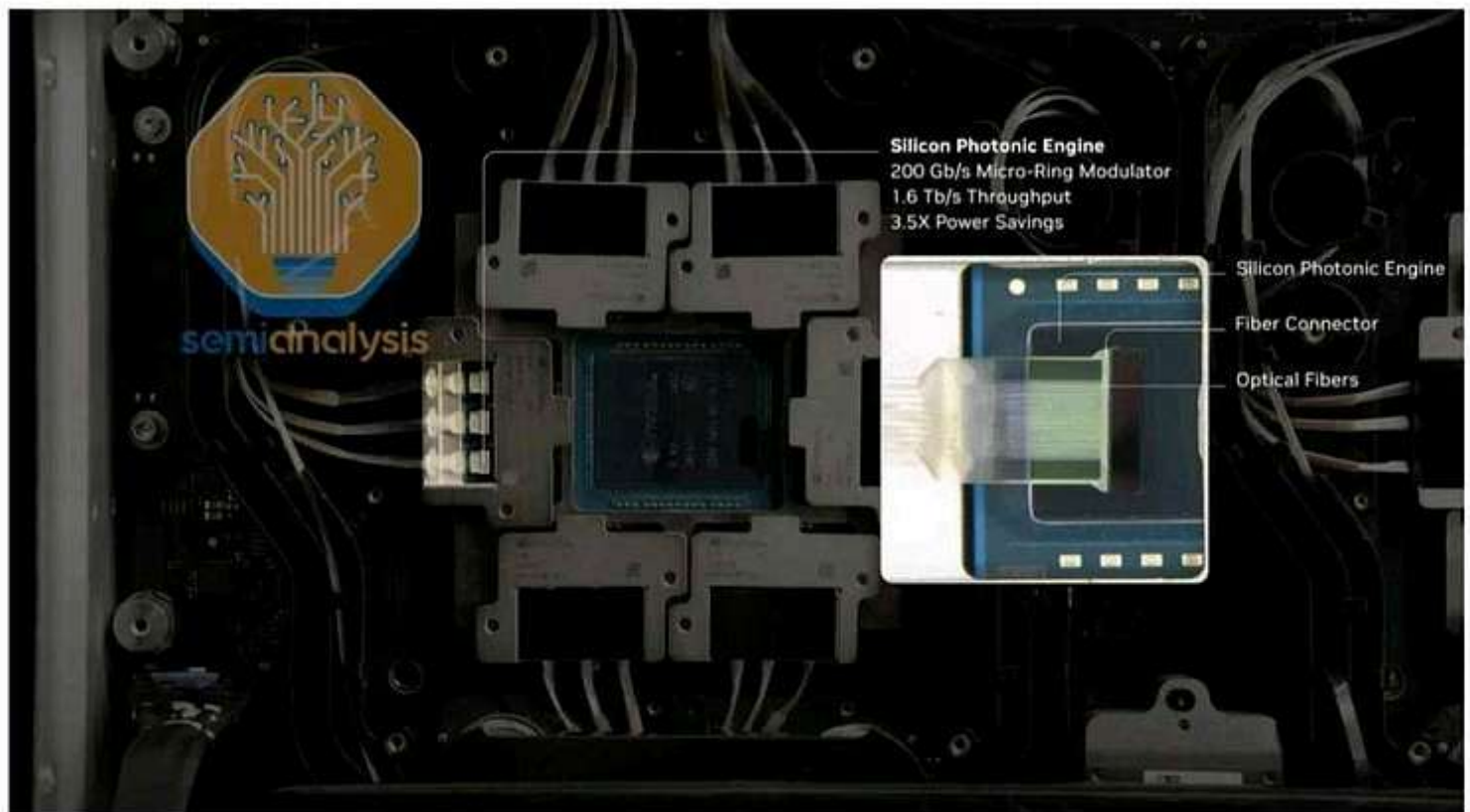


Part 5: Nvidia's CPO Supply Chain

We have discussed the role key components play in CPO systems, in this section we will discuss specific companies in the supply chain and component BOM cost with a focus on Nvidia's suppliers, naming key suppliers for Laser Sources, ELS Modules, FAU, FAU Alight Tools, FAU Assembly, Shuffle Box, MPO Connectors, MT Ferrules, Fibers, and E/O Testing.

Optical Engines

Nvidia's X800-Q3450 CPO switch features 115.2T total throughput and is designed for use with the scale-out network. The initial version will use 72 optical engines each operating at 1.6Tbit/s; later versions will likely transition to 36 optical engines at 3.2Tbit/s each, costing ~\$1,000 per unit (including FAUs). As a result, the total BOM cost for the optical engines amounts to around \$35-40k (for the 3.2T OE version).



Source: Nvidia

External Laser Source (ELS)

The Nvidia X800-Q3450 CPO switch uses 18 ELS modules for laser sources, with each containing 8 Continuous Wave (CW) DFB laser chips. CPO systems need to use

relatively higher-power laser sources, with each CWDFB chip delivering ~350mW of power.

Key industry players capable of producing CW laser units include Broadcom (U.S.), Furukawa (Japan), Lumentum (U.S.), Coherent (U.S.), Yuanjie (China), and Shijia (China). Lumentum, Coherent, Furukawa and Broadcom usually charge higher prices than Chinese vendors (Yuanjie and Shijia). We expect Lumentum to be the sole supplier for Nvidia's initial batches of CPO switch shipments, and Coherent may enter as the second supplier in late 2026. Chinese manufacturers could see opportunities down the road, as CW laser sources are considered relatively standardized and commoditized in general, but there is still somewhat of a moat around building the

Other notable names are Fabrinet (\$FN), TFC Optical (300394.SH), and Foxconn (2354.TW). Fabrinet has long served as the module assembler for Nvidia's in-house optical module units and is now actively building capabilities in OE packaging, testing, and full system assembly. Fabrinet is also viewed as one of the potential candidates for Broadcom's CPO system assembly work, alongside Micas and Foxconn.

TFC has been working closely with Nvidia on CPO design for the past 3-4 years and will be a main supplier for FAUs. The company has been investing in advanced packaging facilities in Suzhou, China, signaling its ambition to secure a larger role in the CPO supply chain.

Electro-optical (E/O) testing equipment: During the testing process, the service vendors mentioned above use electro-optical (E/O) testing tools to ensure system reliability. However, the industry is still developing and has yet to fully converge on a standardized testing methodology for photonic engines. Each vendor is developing various solutions to gain a foothold in this emerging space.

Other key equipment providers in the CPO supply chain include Keysight, Ficontec, Teradyne, Advantest, FormFactor, Chroma, Anritsu, and Multilane. Among them, Keysight is a large player in the field - they are known for providing high-speed testing equipment with premium quality (and premium price!). For example, they [recently announced two new oscilloscopes for 1.6T optical transceiver testing](#). Due to their position in the market, they can well benefit from the CPO trend.

Ficontec has a strong foundation in photonics testing and is now also actively

expanding its electrical testing capabilities. One of its key strengths is wafer-level photonic testing – which improves the efficiency of the traditionally inefficient

photonic testing process. For example, the company [recently introduced a new high-throughput wafer-level testing tool for PICs](#), compatible with existing semi ATE architecture. The machine is claimed as the industry’s first double-sided wafer testing machine. In addition to testing equipment, Ficontec also provides FAU assembly and coupling machines, which we’ll explore in more detail in the next section.

Teradyne is also a large player in the field, with a strong historical presence in electrical testing, but the company has been “very serious” about getting into photonic testing. For example, the company has also recently acquired a startup specializing in packaged optical testing, a move that reflects its strategic push to build out capabilities for CPO.

Chroma has been supplying photonic testing equipment for laser diodes used in 3D sensing and optical communications, and it has the potential to leverage this expertise to enter the CPO space. However, the company’s pace of innovation currently lags some of its competitors.

Senko (Japan), Fukushima (Japan), FOCI (Taiwan), Sumitomo (Japan), and TFC (China). MT ferrules are not particularly difficult to manufacture, but achieving the required precision and robustness still needs a lot of engineering. Companies compete on molding capabilities for producing ferrules that minimize insertion loss during fiber connections.

Among these firms, US Conec also has over three decades of experience developing the technology, and is expected to be one of the major suppliers for Nvidia’s Q3450 CPO systems. Fukushima also has strong mold design and manufacturing capabilities, allowing it to produce high-quality MT ferrules at competitive prices. Meanwhile, FOCI, TFC, and T&S primarily produce MT ferrules for their own in-house FAUs and shuffle boxes – the rationale is to vertically integrate as much of the process as possible to enhance quality control and cost efficiency.

115.2T Nvidia X800-Q3450 CPO Switch Cost and Power (Current Pricing)						
Item	Unit Cost	Quantity	Extended Price	BOM (%)	Power (W)	Extended Power (W)

			Price	(W)	Power (W)
Switch Chip		4		400.0	1,600
Optical Engines		72		7.0	504
FAUs (included in OE)		72		0.0	0
External Laser Source (ELS)		18		8.0	144
MPO connectors and jumper cables		144		0.0	0
Shuffle box		1		0.0	0
Fibers (per meter)		2000		0.0	0
Others					1,300
Total			\$70,640	100%	3,548
Gross Margin			\$134,216	60%	
Selling Price			\$176,600		
3y Service and Warranty			\$28,256		
Selling Price with Service			\$204,856		

Bill of Material estimates are based on current scale of production and could improve as production scales up

Source: SemiAnalysis AI Networking Model

Manufacturing, Assembling, and the Testing Processes

OE fabrication: As discussed above, TSMC will play a crucial role in fabricating the PICs, EICs and integrating them and of the CPO system. Their COUPE platform appears to be the solution of choice for next generation CPO endpoints. Global Foundries, and Tower are the foundries who have strong SiPho capabilities, but their

lack of leading edge CMOS and advanced packaging constrain their ability to provide the higher bandwidth OEs of the future.

Advanced packaging: Outsourced Semiconductor Assembly and Test (OSAT) providers will focus on the back-end processes: including OE packaging, OE testing, and system packaging (laser and coupler integration & testing).

ASE/SPIL (3711.TW), Amkor (\$AMKR), and Shunsin (6451.TW) are primary providers of such solutions. Among them, ASE stands out as a key supplier in Nvidia's supply chain, including involvement in the future Rubin-rack CPO systems, while Shunsin maintains close ties with Broadcom.

T&S Communications (300570.SH) is a leading player in the shuffle box industry. The company has developed automated machines to align fibers within shuffle boxes. The technologies are patented, and it may take time and additional costs for competitors to

technologies are patented, and it may take time and additional costs for competitors to bypass them. T&S's major customer is Corning, and the two companies often work in tandem to serve clients. For instance, Corning helps clients (Nvidia, Broadcom, etc.) to design the fiber networks for their CPO solutions, and subcontract the shuffle box part to T&S.

Molex is another notable player and also entered the market quite early. However, due to lower manufacturing efficiency, its product price is usually ~20% higher than T&S's.



Source: Nvidia

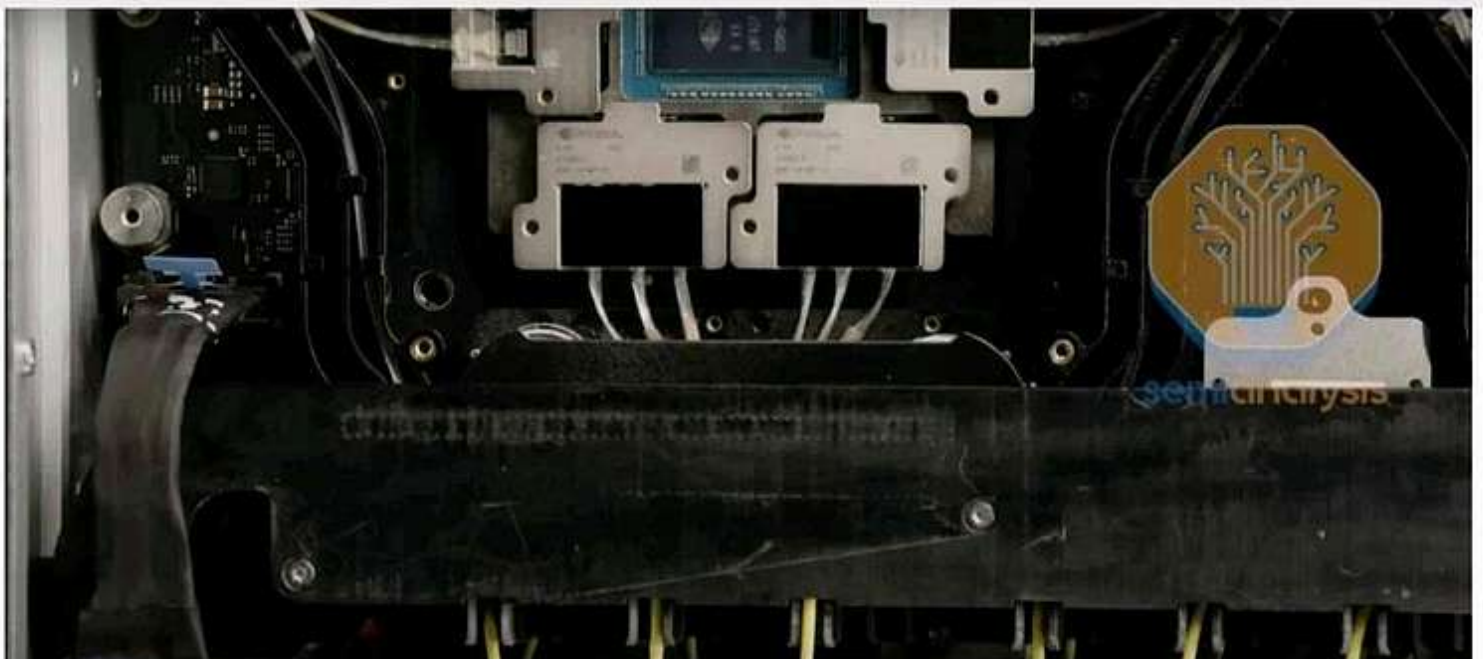
MPO Connectors

On the outskirts of the shuffle box, there are MPO connectors, which connect fibers housed within the shuffle box to an external port. Fiber Optic MPO cables can then plug into this port to connect the switch with other distant switches or to distant NICs. The manufacturing process of MPO connectors primarily involves injection molding, vacuum adhesive filling, and fiber threading of the MT ferrules inside the unit. For the X-800 Q3450 CPO switch, there are 144 MPO connectors needed.

There are multiple players that can produce MPO connectors, including US Conec (U.S.), T&S Communications (China), Senko (Japan), Broadex (China), and Optec (China). They also need to work with the fiber network contract manufacturers (such as Corning) to supply their components into the overall network designs.

MT Ferrules

MT ferrules are critical components used in FAUs, shuffle boxes, as well as MPO connectors. They are used to align multiple fibers in a parallel manner. Many companies can manufacture MT ferrules, including US Conec (U.S.), T&S (China), transmission via the FAU. The company aims to price its machines ~75% lower than comparable Japanese machines, whose price typically charge \$200-250K. The firm expects to have some pile runs in Q4 2025.



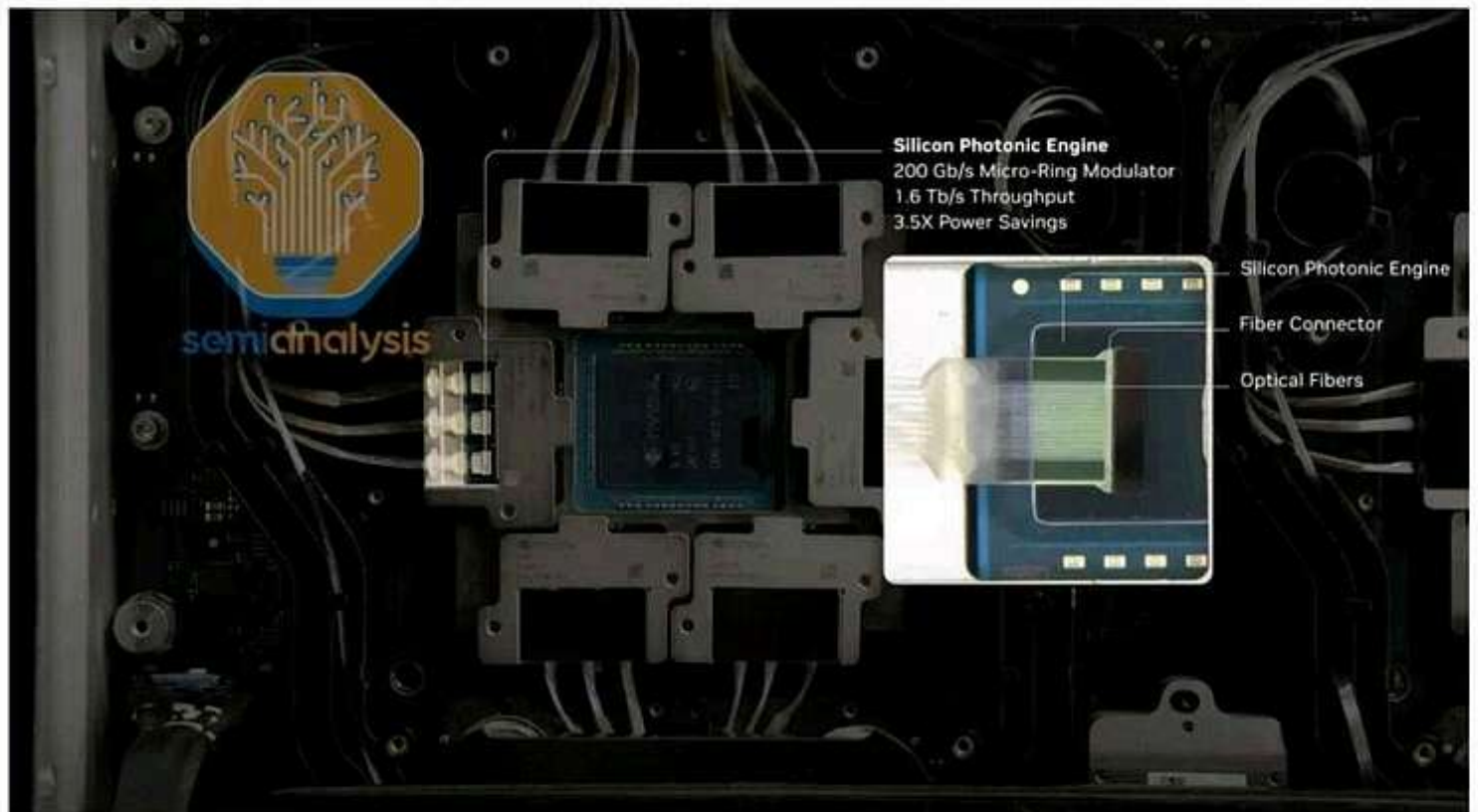


Source: Nvidia

Fiber Shuffle Box

In the case of Nvidia's X800-Q3450 CPO switch, more than 1,000 optical fibers emerge from the OEs, so a shuffle box is needed to organize them and route them to their destinations. Traditionally, this shuffle box alignment process was done manually, with technicians physically arranging the fibers. However, some leading companies have developed automated machines capable of performing the alignment with higher accuracy and efficiency.

The price of a shuffle box is typically linked to the number of fibers it manages. For example, T&S Communications sells a 48-fiber shuffle box for ~\$150, a 300-fiber version for ~\$1,000, and a 500-fiber model for ~\$1,600. For the X800-Q3450 with thousands of fibers, the shuffle box would cost Nvidia more than \$3,000 to procure. The primary BOM components in a shuffle box include MT ferrules and fibers.



Source: Nvidia

The leading companies in FAU are TFC Optical (300394.SH), Senko (9069.JP), and FOCI (3363.TW). TFC has a strong chance of supplying FAUs for the X800-Q3450 CPO switch, while Senko is regarded as the very likely candidate for both the Spectrum X

CPOs and Broadcom's Tomahawk 6 CPO systems. FOCI, meanwhile, is likely to focus more on Nvidia's large-up CPO solutions.

TFC's core strength lies in its robust manufacturing capabilities and access to a large pool of skilled but cost-effective labor in China, a critical factor in maintaining competitiveness, as we discussed in the above paragraph. Additionally, TFC began collaborating with Nvidia on CPO designs around three years ago, and that early partnership is now bearing fruit, with TFC expected to play a key role in Nvidia's ongoing CPO roll-out efforts.

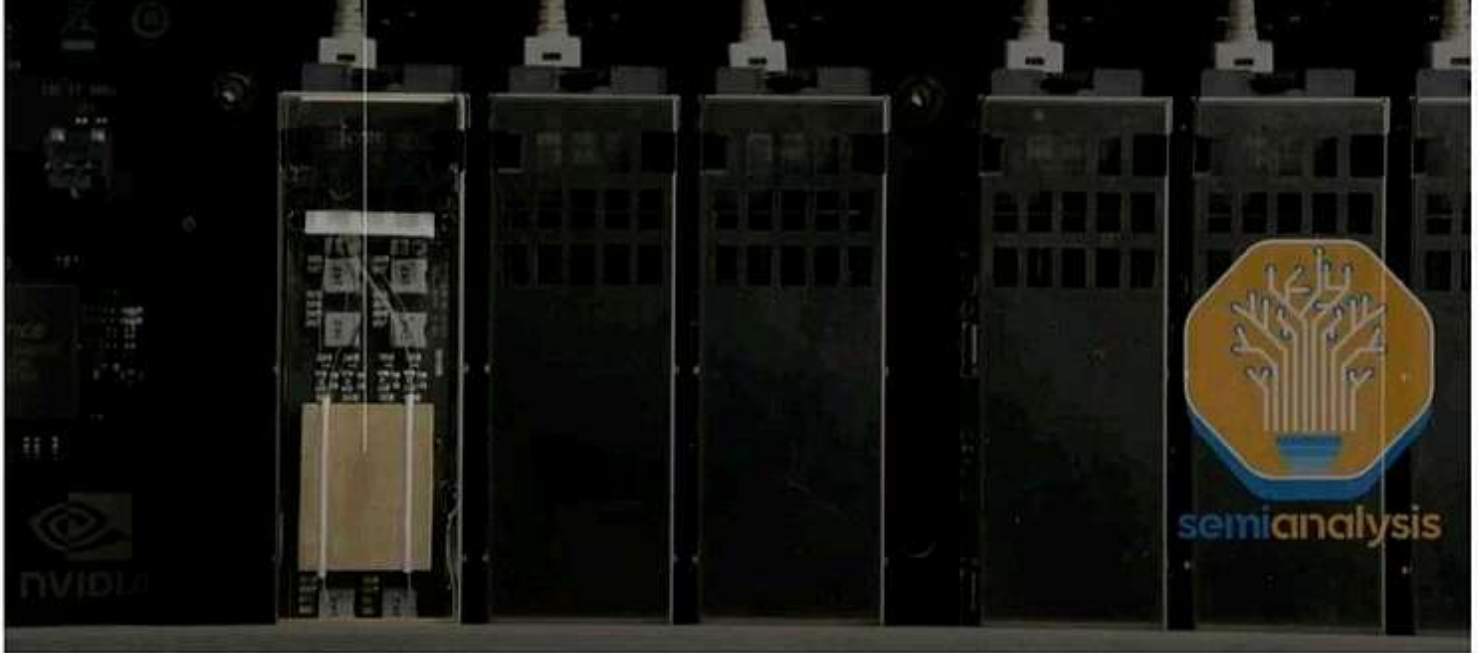
Senko has a signature SEAT (Senko Elastic Averaging Technology) platform where they offer detachable FAU solutions for CPO systems. The company is collaborating closely with GFS on edge-coupling technology, integrating Senko mirrors directly into the wafer trenches. This approach enables wafer-level testing for edge coupling, which has been a key advantage for grating-coupling so far.

Other notable players in the space include Sumitomo, and Advanced Fiber Resources (AFR), among others. AFR is believed to have a close relationship within Broadcom's supply chain.

SiPho chips feature waveguide channels at the micron scale, each requiring precise alignment with the light coming out of the entering the systems. Thus, coupling machines with exceptionally high accuracy are needed during the processes.

FiconTEC (Germany) currently leads the industry in providing high-precision coupling machines. Their machines can be sold at over \$300k per unit; yet they are still highly sought after by clients for their superior accuracy. Meanwhile, All Ring Tech (Taiwan) also provides automated fiber attach equipment. The company has devoted c.10% of its workforce to the space and expects its coupling equipment revenue to gain traction starting in 2026. Lastly, GMT Global (Taiwan) designs FAU bonding, alignment, and inspection machines that correct the wavelength of light to ensure accurate high power laser sources that are required for CPO applications.





Source: Nvidia

Fiber Attach Units (FAUs)

FAUs are crucial passive components responsible for coupling optical fibers with the Optical Engine. High-quality FAUs and their precise alignment are critical to ensure optimal optical performance. One current challenge in the FAU assembly process is that the testing equipment used to measure coupling loss cannot yet be fully automated. As a result, the testing process still relies heavily on manual labor, which

slows down overall production speed and is more costly. Corning estimated an average 10-15 minutes per FAU testing for the Spectrum X CPO systems.

Apart from materials and component costs, labor is a big part of FAU cost – skilled labor is essential to ensure high quality assembly and alignment of fibers in the FAU units. Each 1.6T OE on the X800-Q3450 has an FAU with 20 fibers: 8 transmit, 8 receive and 4 for the external laser. Per system that is a total of 1,440 fibers, including 1152 for Tx/Rx.

Nvidia CPO Supply Chain Market Map					
Laser Source	ELS Module	FAU	FAU Align Tools	FAU Assembly	Shuffle box
LITE	TFC	TFC	finconTEC	Fabrinet	T&S
COHR	O-Net	Senko	All Ring	FOCI	Browave
AVGO	Innolight	FOCI		Foxconn	Molex
Furukawa	Eoptolink	Sumitomo		ASE	
Yamanita					

MPO Connector	MT Ferrule	Fibers	Foundries	OSAT/ Assembly	E/O Testing
US Conec	T&S	Corning	TSMC	Amkor	ficonTEC
Senko	Senko	Sumimoto	Tower	ASE/ SPIL	Keysight
T&S	US Conec	Nittobo	GF	Fabrinet	Teradyne
Browave	Fukushima			TFC	Formfactor
	FOCI				Chroma
	Sumitomo				Multilane

Vendors highlighted green are key players in the sectors & confirmed to be part of Nvidia's CPO supply chain
For more information on market shares and competitive dynamics, contact sales@semianalysis.com.

Source: SemiAnalysis AI Networking Model

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